MILITARY SPECIFICATION

BATTERY, LITHIUM THIONYL CHLORIDE, SINGLE CELL RESERVE

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication of parts, assembling, and packing for a reserve Lithium Thionyl Chloride battery.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance
MIL-T-63420 - Thionyl Chloride Electrolyte

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6135
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MIL-STD-331 - Fuze and Fuze Components Environmental and Performance Test for
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents drawings and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. Army ArmamentResearch, Development and Engineering Center (ARDEC)

PRODUCT AND PACKAGING DRAWINGS:

12550985	Battery Assembly, Single Cell Reserve
12550986	Insulator, Disk
12550987	Pad, Ampule Support
12550988	Shim, Ampule Support
12550989	Dry Room Control
12550995	Terminal Plate Assembly
9275597	Glass
12550996	Plate, Terminal
12550998	Pin, Terminal
12551015	Ampule Assembly, Electrolyte
9317642	Ampule Assembly, Sealing
9328781	Ampule Blank
9317641	Ampule Blank (Anneal)
9328788	Tube, Glass
12551020	Cell Assembly
12551025	Case Assembly, Insulator
9275592	Case
12551007	Insulator, Bottom, Inside
12551027	Insulator, Bottom, Outside
12551028	Separator, Bottom
12441029	Cell Assembly, Anode/Cathode
12551030	Anode Assembly
12551031	Pad, Anode
12551032	Substrate, Anode, Formed
12551033	Substrate, Anode
12551034	Paper, Glass Fiber
12551035	Cathode Assembly
12551036	Substrate, Cathode
12551037	Composition Cathode
12551039	Vertical Separator

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12550752	Batter Primer Activation of M762 Batteries
12550994	Paper Glass Fiber
12550757	Ballistic Environment Test Requirement for M762/M767 Components

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solictation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E-493 - Standard test method of leaks

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

2.3 Order precedence. In the event of a conflict between this specification and references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws or regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Procurement requirements. Procurement of the battery described herein shall be made in accordance with all the requirements stated in this specification and applicable drawings and data listed herein.

3.1.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Temperature range.

3.2.1 Storage temperature range. Batteries manufactured to this specification shall be unaffected by storage temperatures of $-60^{o}F$ to $+160^{o}F$.

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3.2.2 Operating temperature range. Batteries manufactured to this specification shall be capable of meeting full performance requirements from -25°F to +110°F. Degraded performance for certain parameters is permitted from -60°F to -25°F and from +110°F to 155°F. The parameters for degraded performance are shown in the applicable portions of this specification.

3.3 Samples

3.3.1 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3.2 Supplemental samples.

3.3.2.1 Change of qualified product. After qualification, the manufacturer shall notify the purchaser prior to the implementation of any change in design, material, or process which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough dscription of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quantity, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the purchaser for review. Based upon the review, the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested at the manufacturer's expense prior to approval; however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser.

3.4 Electrical performance

3.4.1 Non-activation voltage. Non-activated batteries built to this specification shall have a voltage of less than 5.0 millivolts, in accordance with 4.4.1.1.

3.4.2 Activation and rise time.

3.4.2.1 Activation. Shall be per Procedure 12550752, "Battery Primer Activation of M762 Batteries"

3.4.2.2 Rise time. The rise time of batteries manufactured to this specification shall meet the following requirements:

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Temperature	Max Time to 2.5 Volts
+110° Ambient -25°F -60°F	300 milliseconds 800 milliseconds 1000 milliseconds 2000 milliseconds

3.4.3 Capacity rating. Batteries manufactured to this specification shall be meet the following capacity requirements:

After 15 days with 13K Load:

Temperature	Min Voltage (RL = 13K)
+110°F Ambient -25°F -60°F	3.50 volts 3.40 volts 2.90 volts 2.40 volts

3.4.4 Internal impedance.

3.4.4.1 Internal impedance (Non-activated). The internal impedance of batteries manufactured to this specification shall be not less than 1 megohm. This requirement applies over the temperature range of -60°F to +155°F.

3.4.4.2 Internal impedance (Activated). The internal impedance of batteries manufactured to this specification shall meet the following requirements after 15 days with a 13K load applied:

The minimum voltage with a 15C load applied for 50ms shall be:

Temperature	Min Voltage
+110°F Ambient -25°F -60°F	3.10 volts 3.00 volts 2.20 volts 1.90 volts

3.5 Mechanical

3.5.1 Configuration. Configuration of the batteries manufactured to this specification shall be in accordance with assembly drawing 12550985, Battery Assembly, Single Cell Reserve.

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3.5.2 Leakage. All batteries manufactured to this specification shall be gross leak tested using a bubble test method. All batteries shall also be fine leak tested to 5 X 10 -7 cc of Helium per second. Prior to these leak tests, all batteries shall be temperature cycled with 5 -60°F to +155°F thermal cycles per MIL-STD-202, method 107.

3.5.3 Ampul heat soak. The ampul assembly, electrolyte (12551015), shall withstand two sequential 24 hour heat soaks at 165°F ± 5°F without leakage or breakage, per paragraph 4.4.7.

3.6 Material and finish. The materials and finish for batteries manufactured to this specification shall be in accordance with the drawing 12550985. This shall include all requirements on sub-assembly and piece part drawings which are included in drawing 12550985.

3.7 Environmental and ballistic tests

3.7.1 Environmental tests (Pre-activation). Batteries produced to this specification shall not be activated by a 1.5 meter (5 foot) drop test as described in 4.4.5.1.

3.7.2 Ballistics tests (Post activation). Batteries procured to this specification shall exhibit continuous output for the first 50 milliseconds of flight when exposed to the following conditions when assembled in an M762 fuze:

Set Back Acceleration	30,000 G's
Set Forward Shock	2,500 G's for 3 MSEC
Sideslap Shock	2,500 G's for 0.5 MSEC
Spin	30,000 RPM at center line, offset of 0.35 In.

3.8 Marking. Batteries shall be marked per paragraph 4.5.

3.9 Radiographic examination. Radiographic (x-ray film) batteries manufactured to this specification shall be performed in accordance with paragraph 4.4.2.

3.10 Workmanship. The batteries produced to this specification shall be fabricated and finished in a through workmanlike manner and all manufacturing and assembly operations shall be correctly performed. Metal parts and welds shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, clipped edges, dents, flash, dirt, grease, rust, corrosion products, deposits and all foreign matter or other defects.

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3.10.1 Cleaning. After fabrication, the assembled battery shall be cleaned of smudges, loose or spattered excess weld metal, metal chips and burrs and any other foreign material which may detract from the intended operation, function or appearance of the battery. All corrosive material shall be removed. The cleaning method used shall not be injurious to any part of the assembly nor shall the part be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quallity program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

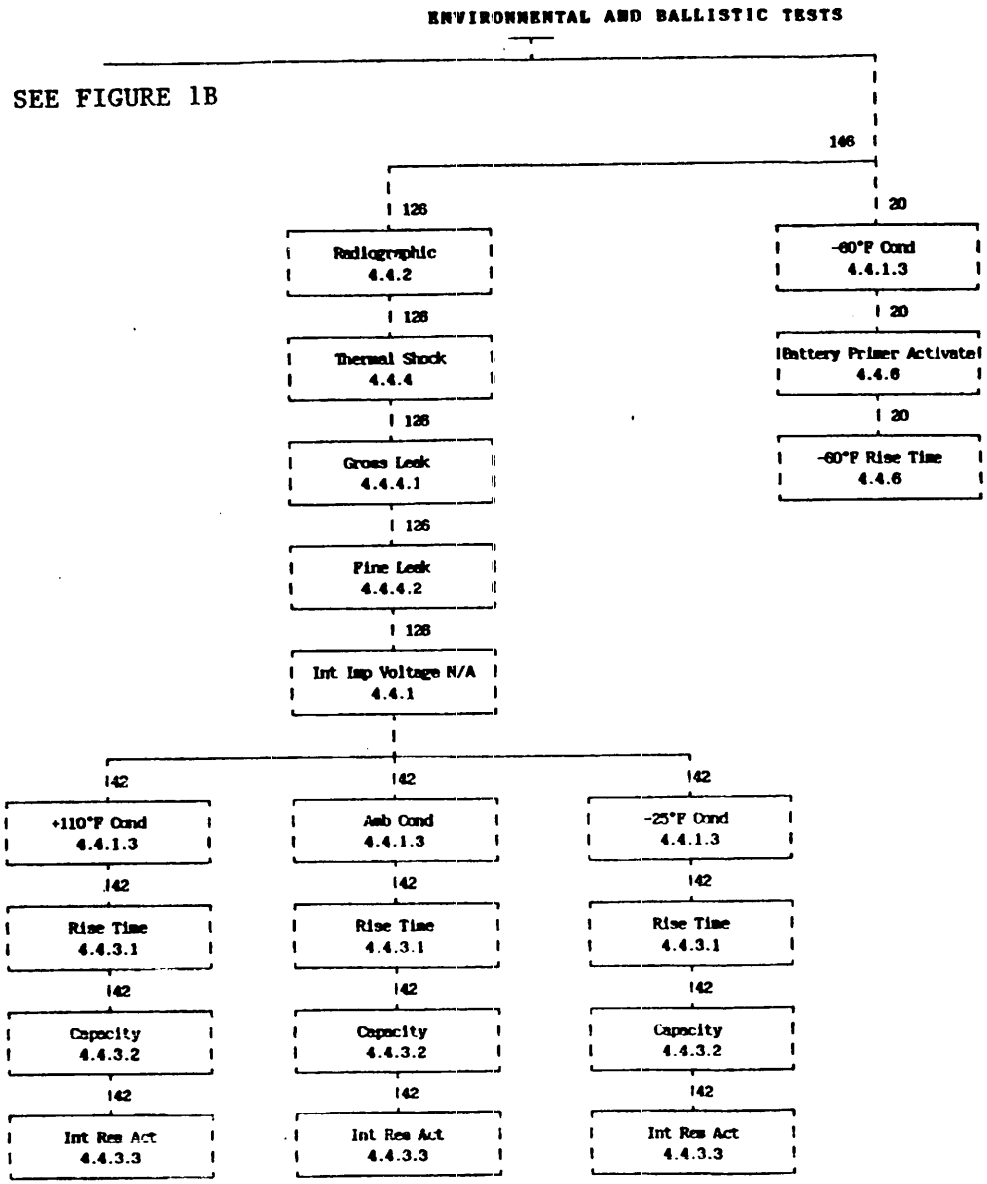
4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a qualification sample as designated by the contracting officer for evaluation in accordance with provisions of 4.2.2. The First Article sample shall consist of two hundred and sixty batteries (260), ten (10) sets of each piece part and assembly listed on the Classification of Defects (C of D's) using the methods and processes establish for quantity production.

4.2.2 Inspection to be performed. See MIL-A-48078 and Table I and Figures 1A, 1B as specified herein.

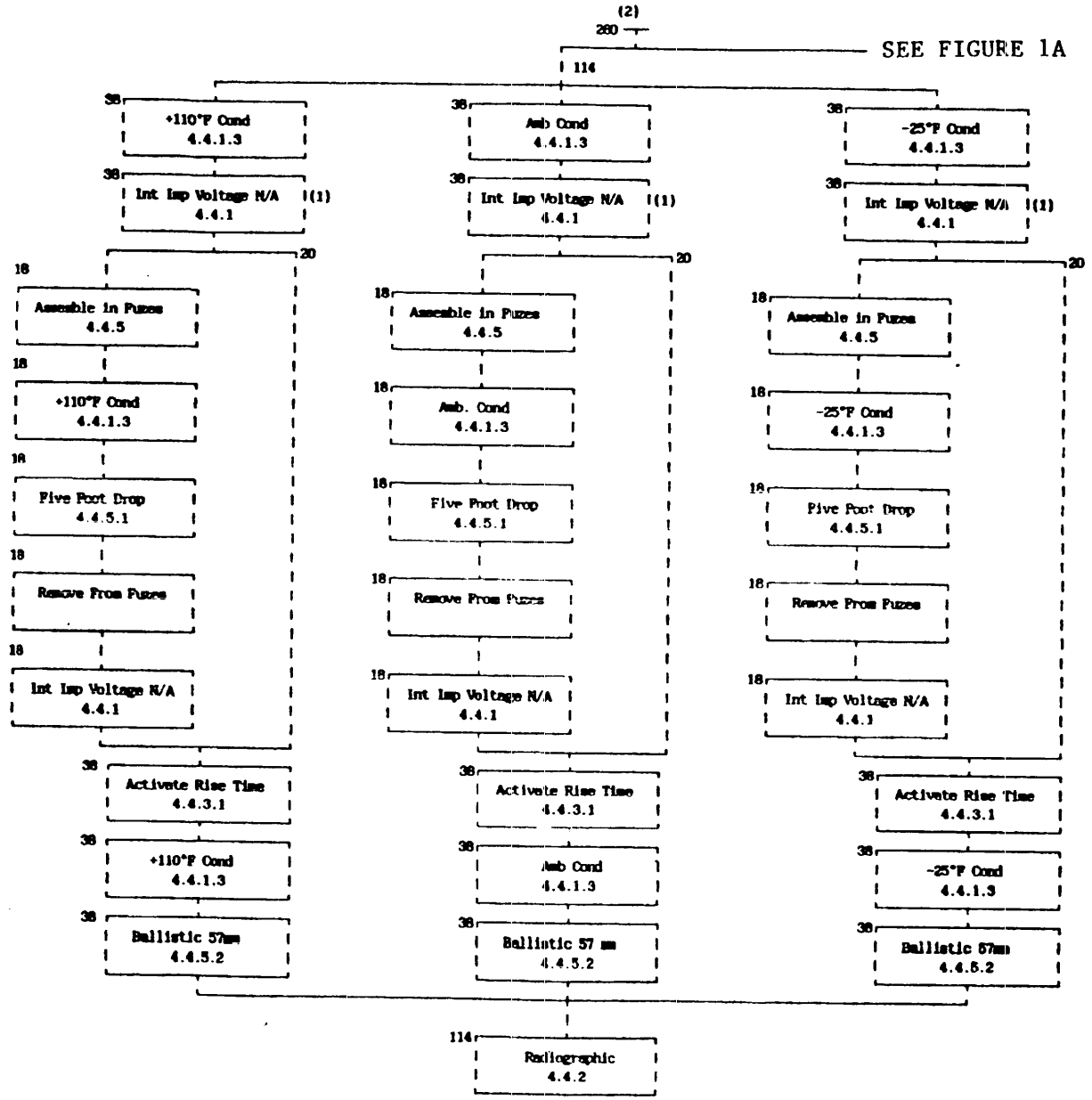
4.2.3 Rejection. See MIL-A-48078.

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RESERVE BATTERY, M762/M767
FIRST ARTICLE ACCEPTANCE TEST FLOW
FIGU 1A

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NOTES:
(1) N/A = Not Activated
(2) Number of Batteries

RESERVE BATTERY, M762/M767
FIRST ARTICLE ACCEPTANCE TEST FLOW
FIGURE 1B

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550985
	Battery, single cell, reserve				NEXT HIGHER ASSEMBLY NONE
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Environmental and Ballistic Tests	54	N = 54 C = 0	3.7	4.4.5
	Workmanship		100%	3.11	3.8
	Radiographic Test	126	100%	3.9	4.4.2
	Thermal Shock	126	100%	3.5.2	4.4.4
	Fine Leak	126	100%	3.5.2	4.4.4.2
	Gross Leak	126	100%	3.5.2	4.4.4.1
	Non-activation Voltage	126	100%	3.4.1	4.4.1
	Risetime, Capacity, Internal Impedance Tests	126	126 C = 0	3.4	4.4.3
	Battery Primer Activation	20	C = 0	3.4.2	4.4.6

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.3 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Sampling Lot Acceptance Testing; and

b. Individual Unit Inspections

4.3.1 Sampling lot acceptance testing

4.3.1.1 Inspection. Inspection shall be as specified in 4.3.2.1, and Figure 2. Inspection may be made at a higher assembly level provided that equal quality assurance provisions are satisfied but not at a lower level than specified.

4.3.1.2 Lot formation. A lot shall consist of devices produced under continuous production at one manufacturing location, in accordance with stabilized methods or techniques, and to the same drawing and specification revision.

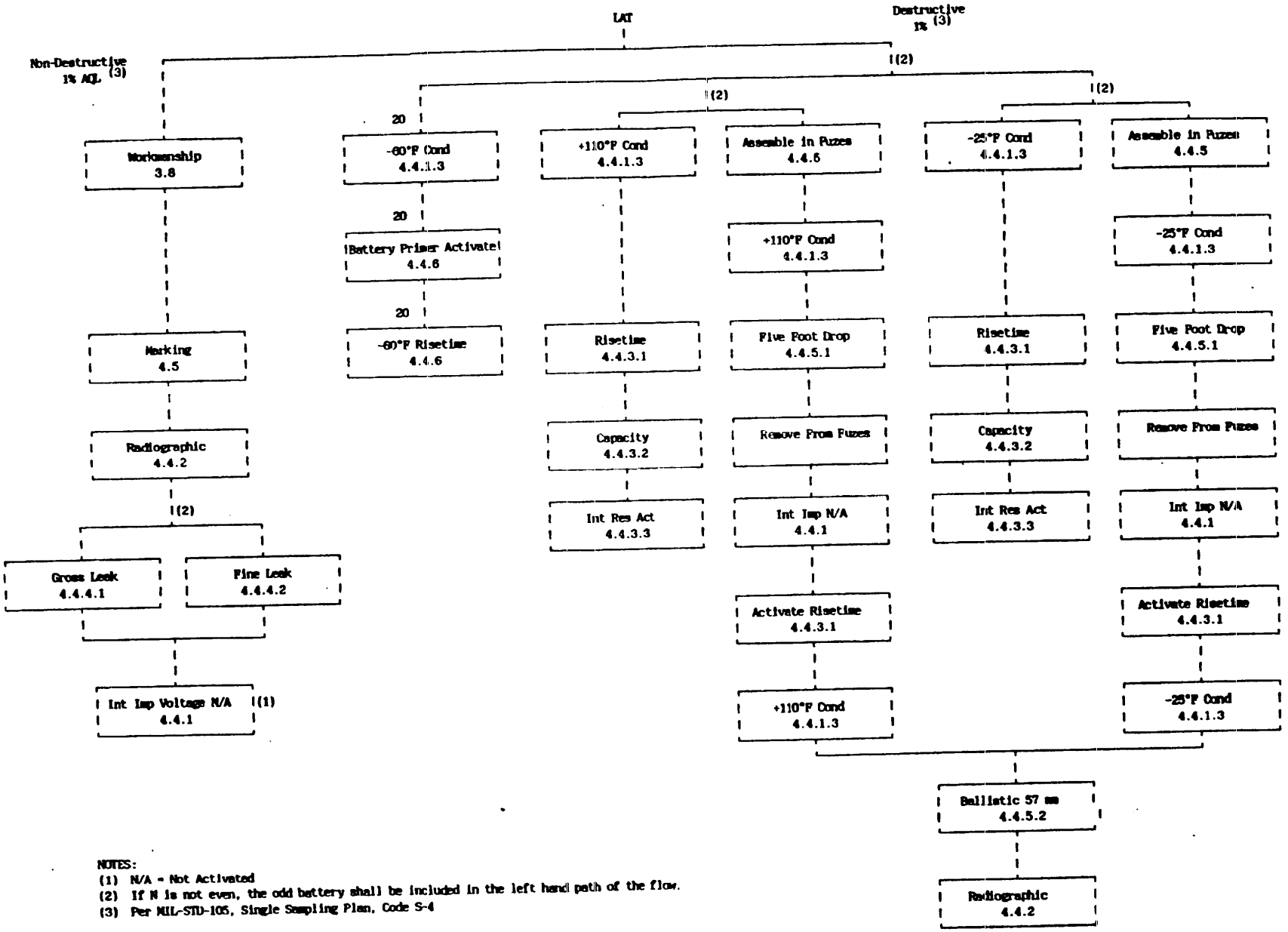
4.3.1.3 Sampling. Sampling shall be in accordance with MIL-STD-105 and shall satisfy an AQL of 1.0%. Unless otherwise specified, inspectin level S-2 shall apply.

4.3.1.4 Lot acceptance. Lots shall be accepted on the basis of the testing identified in 4.3.2.1. In the event the AQL requirements are not satisfied with random samples selected from the lot, a 100% screening of the lot is permitted to remove items with the undesirable characteristics. The screening technique must be non-destructive and acceptable to the purchaser.

4.3.2 Individual unit inspections.

a. Sampling plans. Unless otherwise specified in the classification of defects, sampling plans for all defects shall be in accordance with MIL-STD-105, Inspection Level II, with AQL as shown in the tables.

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NOTES:
(1) N/A = Not Activated
(2) If N is not even, the odd battery shall be included in the left hand path of the flow.
(3) Per MIL-STD-105, Single Sampling Plan, Code S-4

RESERVE BATTERY, M762/M767
LOT ACCEPTANCE FLOW
FIGURE 2

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LOT ACCEPTANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12550985
4.3.2.1	Battery, single cell, reserve		NEXT HIGHER ASSEMBLY NONE

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak Test		1% (NOTE 1)	3.5.2	4.4.4
102	Non-activation Voltage and Resistance		1% (NOTE 1)	3.4.1	4.4.1
103	Risetime, Capacity, Internal Impedance Test		1% (NOTE 1)	3.4.2.2	4.4.3
104	Battery Primer Activation	20	N = 20 C = 0	3.4.2	4.4.7
105	Marking		1%	3.9	4.5
106	Radiographic Examination		1%	3.10	4.4.2
107	Environmental and Ballistic Tests (NOTE 2)		1%	3.7	4.4.5
108	Physical Dimensions		1%	3.5.1	Dwg. 12550985
Minor					
201	Evidence of poor workmanship		1%	3.10	Visual

NOTES: 1. MIL-STD-105, Inspection Level S-2. 2. AQL's per MIL-STD-105, Inspection Level 5-4. 3. Performed by Government.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550985
4.3.2.2	Battery, single cell, reserve				NEXT HIGHER ASSEMBLY NONE
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Shim, Ampul Support Missing		0.40%	3.5.1	Visual
102	Disc, Insulator Missing		0.40%	3.5.1	Visual
103	Ampul Support Pads Missing		0.40%	3.5.1	Visual
104	Weld Integrity - 2 Places		0.40%	3.5.1	Visual
Minor					
201	Evidence of Poor Workmanship		0.65%	3.10	Visual
202	Max Diameter - 2 Places		0.65%	3.5.1	Gage
203	Max Length		0.65%	3.5.1	Gage
204	Marking		0.65%	3.5.1	4.5

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.3.2.3	Disc, insulator	1	1	12550986
				NEXT HIGHER ASSEMBLY: 12550985

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certification of Conformance
Minor					
201	Outside Diameter (Max)		0.65%	3.5.1	Gage
202	Inside Diameter (Min)		0.65%	3.5.1	Gage
203	True Position of ID with OD		0.65%	3.5.1	Gage
204	Thickness Improper		0.65%	3.5.1	Gage
205	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.3.2.4	Pad, ampul support	1	1	12550987
				NEXT HIGHER ASSEMBLY: 12550985

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certificate of Conformance
Minor					
201	Inside Diameter Improper		0.65%	3.5.1	Gage
202	Outside Diameter Improper		0.65%	3.5.1	Gage
203	Thickness		0.65%	3.5.1	Gage
204	True Position of OD		0.65%	3.5.1	Gage
205	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.3.2.5	Shim, ampul support		1 1	12550988
				NEXT HIGHER ASSEMBLY: 12550985
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certificate of Conformance
Minor					
201	True Positions of the 6 Diameters		0.65%	3.5.1	Gage
202	Measurements of 6 Diameters		0.65%	3.5.1	Gage
203	Total Height		0.65%	3.5.1	Gage
204	Evidence of Poor Workmanship		0.65%	3.10	Visual
205	Slot Height and Width		0.65%	3.5.1	Gage

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550995
4.3.2.6	Termianl plate assembly				NEXT HIGHER ASSEMBLY 12550985
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pin Protrusion, Both		0.40%	3.5.1	Gage
102	Leak Rate		0.40%	3.5.1	4.4.4.2
Minor					
201	Pin True Position		0.65%	3.5.1	Gage
202	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550996
4.3.2.7	Plate, terminal				NEXT HIGHER ASSEMBLY 12550995
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall Thickness		0.40%	3.5.1	Gage
Minor					
201	Outside Diameter Improper		0.65%	3.5.1	Gage
202	Evidence of Poor Workmanship		0.65%	3.10	Visual
203	ID and counterbore True Position with OD		0.65%	3.5.1	Gage
204	Taper not present		0.65%	3.5.1	Gage

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550998
4.3.2.8	Pin, terminal				NEXT HIGHER ASSEMBLY 12550995
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certificate of Conformance
Minor					
201	Diameter Improper		0.65%	3.5.1	Gage
202	Evidence of Poor Workmanship		0.65%	3.10	Visual
203	Length		0.65%	3.5.1	Gage

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551007
4.3.2.9	Insulator, bottom, inside				NEXT HIGHER ASSEMBLY 12551025
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Material Improper			3.5.1	Certificate of Conformance
Minor 201	Diameter Improper		0.65%	3.5.1	Gage
202	Thickness Improper		0.65%	3.5.1	Gage
203	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.10	Ampule assembly, electrolyte	1 OF 1	12551015
			NEXT HIGHER ASSEMBLY: 12550985

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper		100%	3.5.1	4.4.8
102	Overall Length (Max)		0.40%	3.5.1	Gage
103	Diameter (Max)		0.40%	3.5.1	Gage
104	Quantity of Electrolyte		0.40%	3.5.1	4.4.9
105	Electrolyte Material Improper		0.40%	3.5.1	Certificate of Conformance
Minor					
201	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551020
4.3.2.11	Cell assemble				NEXT HIGHER ASSEMBLY 12550985
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Evidence of Poor Workmanship		0.65%	3.10	Visual
202	Opening of Anode Assembly Aligned with opening of Cathode		0.65%	3.5.1	Visual

NOTES:

1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551025
4.3.2.12	Case assembly, insulator				NEXT HIGHER ASSEMBLY 12551020
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor 201	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:

1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551027
4.3.2.13	Insulator, bottom, outside				NEXT HIGHER ASSEMBLY 12551025
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certification of Conformance
Minor					
201	Outside Diameter		0.65%	3.5.1	Gage
202	Inside Diameter		0.65%	3.5.1	Gage
203	True Position of ID with OD		0.65%	3.5.1	Gage
204	True Positions of 8 Slots		0.65%	3.5.1	Gage
205	Evidence of Poor Workmanship		0.65%	3.10	Visual
206	Overall Thickness		0.65%	3.5.1	Gage
207	Slot Depths		0.65%	3.5.1	Gage

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551028
4.3.2.14	Separator, bottom				NEXT HIGHER ASSEMBLY 12551025
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Material Improper			3.5.1	Certificate of Conformance
Minor 201	Diameter Improper		0.65%	3.5.1	Gage
202	Thickness Improper		0.65%	3.5.1	Gage
203	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551029
4.3.2.15	Cell assembly, anode/cathode				NEXT HIGHER ASSEMBLY 12551020
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Vertical Separator Presence		0.40%	3.5.1	Visual
102	Overall Diameter		0.40%	3.5.1	Gage
Minor					
201	Anode Assembly and Vertical Separator Relative Position to Cathode Assembly		0.65%	3.5.1	Gage
202	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551030
4.3.2.16	Anode assembly				NEXT HIGHER ASSEMBLY 12551029
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Anode Substrate Note Exposed in Diagonal Slot of Anode Pad		0.40%	3.5.1	Visual
102	Anode Pad Penetrating Perforations in Anode Substrate		0.40%	3.5.1	Visual
103	Anode Pad Not Adhered to Anode Substrate on 100% of Anode Pad Surface		0.40%	3.5.1	Visual
104	Anode Pad not Flush to .010 In Abode Bottom of Anode Substrate		0.40%	3.5.1	Gage
105	Anode Pad Material			3.5.1	Certificate of Conformance
Minor					
201	Length of Cell Body		0.65%	3.5.1	Gage
202	Diameter Improper		0.65%	3.5.1	Gage
203	Spiral Gap in Anode Pad		0.65%	3.5.1	Gage
204	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:

1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551032
4.3.2.17	Substrate, anode, formed				NEXT HIGHER ASSEMBLY 12551030
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weld Integrity Improper		0.40%	3.5.1	Visual
Minor					
201	Inside Diameter Improper		0.65%	3.5.1	Gage
202	Outside Diameter Improper		0.65%	3.5.1	Gage
203	Matchup of Lap		0.65%	3.5.1	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.18	Screen assembly, anode	1 OF 1	12551033
			NEXT HIGHER ASSEMBLY 12551032

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certification of Conformance
Minor					
201	Length Improper A		0.65%	3.5.1	Gage
202	Width Improper B		0.65%	3.5.1	Gage
203	Bottom Border C		0.65%	3.5.1	Gage
204	Screen Hole Pattern D		0.65%	3.5.1	Gage
205	Step In Lead E		0.65%	3.5.1	Gage
206	Evidence of Poor Workmanship		0.65%	3.10	Visual
207	Overall Thickness F		0.65%	3.5.1	Gage

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.3.2.19	Cathode assembly		1 1	12551035
				NEXT HIGHER ASSEMBLY
				12551029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Composition Cathode Not to Extend Beyond Edges of Cathode Substrate		0.40%	3.5.1	Visual
202	Assembly Thickness		0.65%	3.5.1	Gage
203	Evidence of Poor Workmanship		0.65%	3.8	Visual
204	Composition Cathode to Edge of Substrate Dimension		0.65%	3.5.1	Visual
205	Overall Thickness		0.65%	3.5.1	Gage
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1. All AQL's per MIL-STD-105, Inspection Code Letter II

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551036
4.3.2.20	Substrate, cathode				NEXT HIGHER ASSEMBLY 12551035
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material Improper			3.5.1	Certificate of Conformance
Minor					
201	Length Improper A		0.65%	3.5.1	Gage
202	Width Improper B		0.65%	3.5.1	Gage
203	Bottom Border C		0.65%	3.5.1	Gage
204	Screen Hole Pattern D		0.65%	3.5.1	Gage
205	Evidence of Poor Workmanship		0.65%	3.10	Visual
206	Step in Leads E		0.65%	3.5.1	Gage
207	Overall Thickness F		0.65%	3.5.1	Gage

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12551037
4.3.2.21	Cathode				NEXT HIGHER ASSEMBLY 12551035
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight Improper		0.40%	3.5.1	Gage
102	Thickness Improper		0.40%	3.5.1	Gage
103	Material Improper			3.5.1	Certification of Compliance
104	Percentage of Acetylene Black		0.40%	3.5.1	Gage
105	Percentage of PTFE		0.40%	3.5.1	Gage
Minor					
201	Length Improper		0.65%	3.5.1	Gage
202	Width Improper		0.65%	3.5.1	Gage
203	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.3.2.22	Separator, vertical	1	1	12551039
				NEXT HIGHER ASSEMBLY: 12551029

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Materials Improper			3.5.1	Certificate of Conformance
102	Grain Direction		0.40%	3.5.1	Visual
Minor					
201	Length Improper		0.65%	3.5.1	Gage
202	Width Improper		0.65%	3.5.1	Gage
203	Thickness Improper		0.65%	3.5.1	Gage
204	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70824 (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.3.2.23	Case		1 1		9275592
					NEXT HIGHER ASSEMBLY
					12551025
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Materials Improper			3.5.1	Certificate of Conformance
102	Hardness Improper	3		3.5.1	3.6
103	Inside Diameter Improper		0.40%	3.5.1	Gage
104	Length Improper		0.40%	3.5.1	Gage
105	Thickness Improper		0.40%	3.5.1	Gage
Minor					
201	Finish Improper		0.65%	3.5.1	Visual
202	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES:
1. All AQL's per MIL-STD-105, Inspection Code Letter II

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9328781
4.3.2.24	Ampul blank				NEXT HIGHER ASSEMBLY 12551015
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ampul Diameter Improper		0.40%	3.5.1	Gage
102	Ampul Length to Shoulder Improper		0.40%	3.5.1	Gage
103	Stricture OD Improper (2 Places)		0.40%	3.5.1	Gage
104	Stricture ID Improper		0.40%	3.5.1	Gage
105	Stricture True Position Improper		0.40%	3.5.1	Gage
106	Wall Thickness Improper		0.40%	3.5.1	Gage
107	Bottom Thickness Improper		0.40%	3.5.1	Gage
108	Material Improper			3.5.1	Certificate of Conformance
Minor					
201	Overall Length		0.65%	3.5.1	Gage
202	Evidence of Poor Workmanship		0.65%	3.10	Visual

NOTES: 1. AQL's per MIL-STD-105, Inspection Code Level II

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4.4 Test methods

4.4.1 Non-activation impedance and voltage

4.4.1.1 Voltage (Non-activated). Batteries in this specification shall be measured for non-activation voltage using an acceptable digital millivolt meter, as approved by the government having a minimum input impedance of 10^6ohms with 500K ohms across the cell. Any battery which measures more than five (5.0) millivolts shall be classed defective and removed from further tests.

4.4.1.2 Internal impedance (Non-activated). The internal impedance of a non-activated battery shall be measured with digital volt-ohm meter or other suitable instrument. The non-activated batteries built to this specification shall have a measured impedance of not less than one megohm.

4.4.1.3 Preconditioning. Depending on the test flows of Figures 1A, 1B or Figure 2, the non-activated voltage and internal impedance tests may be preceded by temperature conditioning. Where called for, the batteries shall be conditioned to the test temperature specified in Figures 1A, 1B or Figure 2 for a minimum of two hours and held to a temperature accuracy of $\pm$ 5°F. Testing shall be performed within 2 minutes after removal from the conditioning chamber.

4.4.2 Radiographic examination. Inspection equipment and image quality of the exposed x-ray films for the examination shall be satisfactory to the purchaser. The direction of the x-ray beam employed in the inspection shall be prependicular to the cylindrical wall of the battery, with the anode lead to screen welds positioned to the right. Retain x-ray films in the file during production. (Non-destructive test).

X-ray negatives of the batteries shall be examined for the following:

a. There shall be continuity of the anode lead.

b. There shall be no sharp bends in the anode lead.

c. The anode lead shall not be bent against the outside case.

d. There shall be no discernible sharp bends blocking of damage to the cathode leds.

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e. There shall be no discernible damage to the anode or cathod screens.

4.4.3 Risetime, capacity and impedance test. Batteries to be tested shall be preconditioned prior to activation for a minimum of two hours at the test temperatures indicated in the FAAS or Lot Acceptance test flows of Figures 1A, 1B or Figure 2. The test temperature shall be maintained within ± 5°F. The batteries shall remain at the test temperatures for the duration of the testing and not be removed from the test chambers.

4.4.3.1 Risetime test. Following conditioning, each battery to be tested for risetime shall be connected to a 13 Kohm ± 1% load prior to activation. Activation of the battery for this test shall be achieved per procedure 12550752, Risetime, Capacity and Internal Impedance Test for M762 Reserve Cell, Battery Primer Activation. Time shall be measured from immediately after the activation shock until the terminal voltage across the 13 K ohm load reaches 2.50 volts. The risetime shall be measured with a storage oscilloscope or other suitable instrument and recorded. The measured rise time shall not exceed the values shown below for the associated temperature. The test shall be performed at the conditioning temperature.

The maximum measured risetimes shall not exceed the following values:

TEMPERATURE	MAXIMUM RISE TIME TO 2.5 VOLTS
+110°F	300 Milliseconds
Ambient	800 Milliseconds
-25°F	1000 Milliseconds
-60°F	2000 Milliseconds

4.4.3.2 Capacity test. Those batteries tested for rise time per 4.4.3.1 above shall be used for capacity testing. The batteries shall be maintained at the conditioning temperature for 384 +4, -0 hours following activation with the 13K ohm resistive loads attached. The terminal voltage, time, and temperature shall be recorded every 4 hours for the 384 hour period. The terminal voltage shall not fall below the following values for each temperature:

TEMPERATURE	MINIMUM VOLTAGE (RL = 13K)
110°F	3.50 Volts
Ambient	3.40 Volts
-25°F	3.90 Volts
-60°F	2.40 Volts

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4.4.3.3 Internal impedance (Activated). The batteries that were previously tested for capacity 4.4.3.2 shall be tested for internal impedance at the completion of the 384 hour +4, -0 period. A 150 ohm ± 2% resistive load shall be applied for 50 ms ± 1 ms in parallel with 13K ohm ± 1% load for each battery. Mininum battery terminal voltage under load shall be measured and also measured at 5 ms after removal of the load (recovery voltage). The recovery voltage shall be a minimum value of 2.8 for all temperatures. The minimum terminal voltage under load conditions shall be:

TEMPERATURE	TERMINAL VOLTAGE
+110°F	3.10 Volts
Ambient	3.00 Volts
-25°F	2.20 Volts
-60°F	1.90 Volts

4.4.4 Leak test. Prior to leak testing, the batteries shall be subjected to five (5) cycles of thermal shock per MIL-STD-202, Method 107, Condition A, except that the maximum temperatures shall be -60°F and +155°F.

4.4.4.1 Gross leak. Each battery shall be subjected to the gross leak test. The test shall be performed in accordance with MIL-STD-202, Method 112B, Condition B. (Any evidence of bubbles shall be cause for rejecting the gross leak test using the fine leak test equipment of 4.4.4.2, provided the equipment has the capability of gross leak detection). Any battery failing to meet the leak rate requirements shall be rejected.

4.4.4.2 Fine leak. Batteries manufactured to this specification shall be tested in accordance with ASTM-E-493, Method A. The volume for calculating the actual leak rate shall be taken as 0.6cc. The calculated leak rate shall be less than 5×10^{-7} cc per second (helium). Any battery failing this test shall be rejected.

Leak testing as called out in the Classification of Defects and Tests shall be performed per ASTM-E-493, using fixturing appropriate for the tested assembly or part.

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4.4.5 Environmental and ballistic testing.

4.4.5.1 1.5 Meter (5 foot) drop test. Batteries to be drop tested shall be preconditioned prior to drop testing at the temperatures shown in Figures 1A or Figure 2. The preconditioning temperatures shall be maintained at ± 5°F for a minimum of 4.0 hours. The batteries shall be assembled into M762 fuze assemblies prior to temperature conditions. The assemblies need not be operational, but shall be mechanically representative of the fuze.

After temperature conditioning the fuze assemblies shall be subjected to a 1.5 meter drop test in accordance with MIL-STD-331, Test 111.1, Procedure 2, Condition A with the exception that less than 10 batteries may be dropped. The elapsed time between removal from the conditioning chamber and the start of the drop test shall not exceed 15 minutes. The batteries shall be monitored for 24 hours following the drop test. To assure that no battery was activated by this test, monitoring shall be performed with a voltage recording system with a minimum input impedance of 10^6ohms. A 500K ohm ± 5% resistive load shall be across the battery during monitoring. Any battery which measures greater than 5 millivolts (5.0 MV) shall be failed and considered activated.

4.4.5.2 Ballistic testing. This test is performed by the government on batteries that are activated at ambient temperature per 4.4.6. Batteries shall reach 2.50 volts within 800 ms of activation and reach a minimum of 3.50 volts within 2 seconds after activation with a 13K ohm ± 1% load attached to the battery terminals.

Following activation the batteries shall be mounted in a T-can projectile (inert) with a weight of 6.36 pounds. The projectile shall be exposed for a minimum of four hours at the conditioning temperatures specified in Figure 1A or Figure 2.

The projectile shall contain circuitry to continuously monitor battery terminal voltage for 50 milliseconds ± 1.0 milliseconds after setback and detect voltage interruptions greater than 5.0 microseconds ± 10%.

The projectile shall be fired in a vertical recovery test with a 57MM type M1 weapon. The propellant shall be type MP-MI (RAD 68308), with a nominal 41 ounce quantity. The quantity may be adjusted according to ambient conditions.

A voltage interruption of more than 5.0 ± 10% microseconds during the first 50 milliseconds following setback shall be considered a battery failure.

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4.4.6 Battery primer activation. Batteries to be activated using battery primers shall be tested according to 12550752, 4.0, and 5.1 through 5.5, Risetime, Capacity, Internal Impedance Test Procedure for M762 Reserve Cell-Battery Primer Activation.

The maximum measured risetimes shall not exceed the following values:

TEMPERATURE	MAXIMUM RISETIME TO 2.5 VOLTS
+110°F	300 Milliseconds
Ambient	800 Milliseconds
-25°F	1000 Milliseconds
-60°F	2000 Milliseconds

4.4.7 Ampul heat soak. Each ampul assembly (12551015) shall be subjected to two consecutive 165°F ± 5°F heat soaks, each soak for a minimum of 24 hours. After each soak the ampules shall be examined under 50x magnification for signs of leakage, cracks, or capillary passage in the ampul nipple. Any parts with these defects shall be removed from the lot.

4.4.8 Electrolyte fill. The ampules shall be filled with .53cc - .4cc Thionyl Chloride Electrolyte, specification MIL-T-63420.

4.5 Marking. See Drawing 12550985. Batteries procured to this specification shall be marked with a polarity mark (+) and the lot/date code. In addition, the minimum protective packaging shall show the part number and be marked with "WARNING: EXCESSIVE HEATING OF THE BATTERY MAY CAUSE A RUNAWAY REACTION CAUSING THE CELL TO OPEN MILDLY (VENT) OR VIOLENTLY (EXPLOSION). DO NOT INCINERATE, SHORT CIRCUIT, USE OR STORE OUTSIDE SPECIFIED TEMPERATURE RANGE, PUNCTURE, CRUSH, DISASSEMBLE, RECHARGE OR FORCE OVER-DISCHARGE (REVERSE)".

5. PACKAGING

5.1 Packaging and preservation. Battery assemblies purchased to this specification shall be packed in accordance with MIL-STD-1169. The batteries must be packaged such that shorting of a cell cannot occur.

6. NOTES

6.1 Intended use. The detonator assembly covered by the specification is intended for use in munition assemblies.

6.2 Ordering data. See Sections 6.1 and 6.2 of MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. (See MIL-A-38078). Submit equipment designs as required to:

Commander
AMCCOM
ATTN: AMSMC-QAR-R(D)
Picatinny Arsenal, New Jersey 07806-5000

This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
AMCCOM
ATTN: AMSMC-QAR-R(D)
Picatinny Arsenal, New Jersey 07806-5000

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Subject term (key word) listing.

Activation
Dry Room Control
Electrolyte
Internal
Impedance
Lithium Thionyl Chloride
Rise Time

Custodian
Army-AR

Preparing activity
Army-AR

(Project 6135-A217)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-70824 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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